

- **Abstract:** Write a concise summary of the problem space, the methods you proposed, and the primary empirical results.

The spread of misinformation through online news platforms has become a significant societal challenge, increasing the need for automated systems capable of identifying fake news articles accurately and efficiently. This project aims to create a machine learning-based fake news detection system using Natural Language Processing (NLP) techniques. A combined dataset of real and fake news articles, provided by Kaggle, was preprocessed through text cleaning and normalization procedures, including lowercasing, punctuation removal, and stop-word filtering. The textual data was transformed into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. A logistic regression classifier was then trained to distinguish between fake and real news content. The model was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Experimental results demonstrated that the proposed approach achieved high classification performance, showing that TF-IDF features combined with logistic regression provide an effective and computationally efficient solution for fake news detection. These findings highlight the potential of machine learning techniques in combating misinformation in digital media environments.

- **Introduction:** Clearly define the problem you are solving, discuss its real-world motivation, and outline the specific contributions your project makes.

The rapid growth of digital media and online news platforms has transformed how people access and consume information. While these technologies provide instant access to global events, they have also contributed to the widespread dissemination of misinformation and fake news. Fake news refers to intentionally misleading or fabricated information presented as legitimate journalism. The spread of false information through social media and online publishing platforms can influence public opinion, damage trust in reliable news sources, and affect important societal processes such as elections, public health decisions, and political discourse. As the volume of online content continues to increase, manually verifying the credibility of news articles has become increasingly difficult, creating a strong need for automated fake news detection systems.

I have a personal stake in the issue of detecting fake news due to my interest in journalism and news. It is important to be updated with what is happening in the world and in America, and it becomes harder to be continually updated when fake news muddies the waters. Many individuals, including myself, have fallen for fake news that was false or twisted the truth, due to not doing further research into the sources of the content. This detector will have a direct impact on sifting through the vast media landscape that allows people outside of credible journalists to public news content.

This project addresses the problem of automatically classifying news articles as either fake or real using machine learning and natural language processing techniques. Because news articles are primarily text-based, NLP methods provide an effective way to analyze linguistic patterns and identify characteristics commonly associated with misinformation. Traditional manual fact-checking methods are time-consuming and require a lot of human effort, where

instead machine learning models can process large amounts of textual data quickly and consistently.

The proposed system uses a TF-IDF vectorization approach to convert news articles and their text into numerical feature representations. These features are then used to train a logistic regression classifier that is capable of distinguishing between fake and real news articles. TF-IDF was selected because it effectively captures the importance of words within documents while reducing the influence of commonly occurring terms. Logistic regression was chosen due to its efficiency, interpretability, and strong performance in text classification tasks.

The primary contributions of this project to the real-world are a few key things: the development of a complete fake news detection pipeline, including data preprocessing, feature extraction, model training, and evaluation; application of TF-IDF vectorization to transform textual news data into machine learning-compatible numerical representations; implementation of a logistic regression classifier for binary classification of fake and real news articles; evaluation of the model using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis; and demonstration that relatively simple and computationally efficient machine learning techniques can achieve strong performance in fake news detection tasks.

The results of this project demonstrate the effectiveness of combining TF-IDF feature extraction with logistic regression for identifying misinformation in news articles. This work highlights the broader potential of machine learning and NLP methods in addressing challenges related to information reliability and digital media trustworthiness.

- **Related Work:** Contextualize your project within existing literature. Group related studies conceptually, such as categorizing prior work into clustering approaches, predictive modeling, and hybrid frameworks.
- **Methodology:** Detail your technical approach. Explicitly describe your data preprocessing steps, including how you handled binary mapping, one-hot encoding, and feature scaling. If proposing architectural changes, outline the theoretical design of your models.

This project utilized the *Fake News Detection Datasets* obtained from Kaggle, which consisted of two comma-separated value files (CSV): *Fake.csv* and *True.csv*. The *Fake.csv* dataset contained news articles labeled with fake news, while the *True.csv* dataset contained legitimate news articles labeled as real news. Each dataset included textual attributes such as article titles and article body content. For this project, the title and text fields were selected as the primary features because they contained the most relevant linguistic information for classification. To prepare the data for supervised machine learning, both datasets were combined into a single dataframe after assigning binary class labels. Fake news articles were mapped to the value 1, while real news articles were mapped to the value 0. This binary mapping allowed the problem to be treated as a binary classification task. After labeling, the datasets were concatenated and randomly shuffled to eliminate ordering bias and improve model generalization.

- **Experiments & Results:** Present your findings systematically. Compare your proposed model against established baselines or conduct ablation studies, utilizing appropriate evaluation metrics such as mean IoU, MSE, or accuracy. Incorporate visualizations like PCA projections, metric graphs, or feature importance plots to support your conclusions.
- **Conclusion:** Summarize your core findings, acknowledge the limitations of your approach, and suggest avenues for future work.